

Roll back

Repair step after the trust detector has already found bad contributors

At every time step k , before the observer update is finalized, we save the information needed to replay the update later.

$$\mathcal{B}(k) = \left\{ \hat{x}_i^{(j)}(k), \hat{x}_0^{(j)}(k), \hat{x}_l^{(j)}(k), w_{i0}^{(j)}(k), w_{il}^{(j)}(k), u_j(k), \Delta t \right\}$$

When a malicious set $\mathcal{M}(t)$ is detected at time t , choose the oldest saved :

$$\hat{x}_{i,\text{rb}}^{(j)}(k_0) = \hat{x}_{i,\text{save}}^{(j)}(k_0) \quad k_0 = t - K$$

Then replay each step $k = k_0, \dots, t - 1$, excluding malicious sources:

$$z_{i,\text{rb}}^{(j)}(k) = \hat{x}_{i,\text{rb}}^{(j)}(k) + \mathbf{1}_{0 \notin \mathcal{M}} w_{i0}^{(j)}(k) \left(\hat{x}_0^{(j)}(k) - \hat{x}_{i,\text{rb}}^{(j)}(k) \right) + \sum_{l \in \mathcal{N}_i \setminus \mathcal{M}} w_{il}^{(j)}(k) \left(\hat{x}_l^{(j)}(k) - \hat{x}_{i,\text{rb}}^{(j)}(k) \right)$$

$$\hat{x}_{i,\text{rb}}^{(j)}(k+1) = f \left(z_{i,\text{rb}}^{(j)}(k), u_j(k), \Delta t \right)$$

Final replacement:

$$\hat{x}_i^{(j)}(t) \leftarrow \hat{x}_{i,\text{rb}}^{(j)}(t)$$